

CHAPTER 1

Introduction to Software Project Management

Outline

- Project Management Fundamentals
- Characteristics of a project: product, budget, schedule trade-off
- Software Project Management Tasks
- Major Problems in Software Projects

What is management?

- It is achieving goals in a way that makes the best use of all resources
- This involves the following activities:
- **Planning** – It is the basic function of management. “Planning is deciding in advance - what to do, when to do & how to do. It bridges the gap from where we are & where we want to be”. “**deciding what is to be done**”
- **Organizing** – It is the process of bringing together physical, financial and human resources and developing productive relationship among them for achievement of organizational goals. “**making arrangements**”
- **Staffing** – It is the function of manning the organization structure and keeping it manned. Manpower, training. “**selecting the right people for the job**”
- **Directing** – Supervision, Motivation, Leadership, Communication. “**giving instructions**”

What is management?- Cont'd

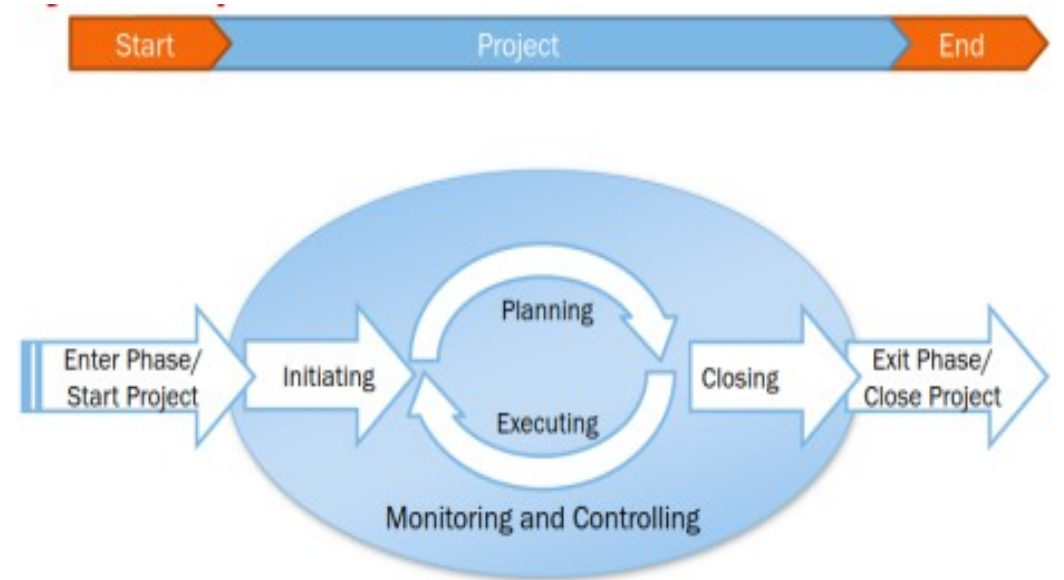
- **Controlling** –The purpose of controlling is to ensure that everything occurs in conformities with the standards.
- **Monitoring** – checking on progress
- **Innovating** – coming up with solutions when problems emerge
- **Representing** – communicating with clients, users, developers and other stakeholders.

What is a project?

- It is a temporary effort to create a unique product or service. Projects usually include constraints and risks regarding cost, schedule or performance outcome.
- It is a planned set of interrelated tasks that are coordinated to achieve a specific objective and to be executed over a fixed period and within certain cost and other limitations.

Project Life cycle

- It is a series of stages that a project goes through from initiation to closure. The project lifecycle typically consists of the following phases:
- **Planning**- deciding what is to be done. This is the most challenging and time consuming step.
- **Execution**- is doing what is planned. This is where the actual project is begins.
- **Monitoring**- checking on progress.
- **Controlling**- taking action to remedy hold-ups.
- **Closing**- Project completion and handover the project to stakeholders.



What is Project Management?

- It is the discipline of planning, organizing, motivating, and controlling resources to achieve specific goals.
- It is a methodical approach to planning and guiding project processes from start to finish.
- **Why Project Management?**
 - Better control of financial, physical, and human resources
 - Improved customer relations
 - Shorter development times
 - Lower costs
 - Higher quality and increased reliability
 - Improved productivity
 - Better internal coordination

Project Manager

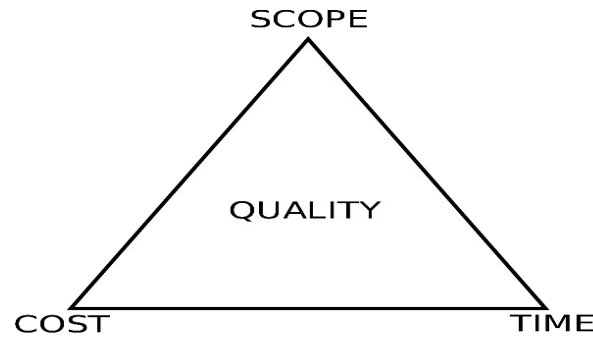
- A person who has a diverse set of skills (e.g. good general skills, technical management, conflict management, customer relationship management and leadership skills), and responsible for:
 - initiation,
 - planning,
 - executing,
 - controlling,
 - monitoring, and
 - closing down a project

Characteristics of Projects

- Non-routine
- Planned
- Aiming at a specific target
- Work carried out for a customer
- Involving several specialisms
- Made up of several different phases
- Constrained by time and resources
- Large and/or complex

Project Trade-off

- Key **project management** responsibilities include
 - creating clear and realistic project objectives,
 - building project requirements,
 - managing the triple constraint for projects, which is cost, time, and scope
- The Triple Constraint
 - The **scope/product** constraint refers to what must be done to produce the project's **end result**.
 - The **time/schedule** constraint refers to the amount of time available to complete a **project**.
 - The **cost/budget** constraint refers to the budgeted amount available for the **project**.
- It is the project manager's duty to balance these three often competing goals.



Project Dimensions

- Effective SPM practices focus on four factors namely people, product, process and Technology. All of them need equal attention in the SPM process.
- **People:** This refers to the human resources involved in a software project, including the skills, knowledge, and expertise of the team members. Effective people management involves recruiting, training, motivating, and supporting team members, as well as managing any conflicts or performance issues.
- **Process:** SPM involves following a process that helps to ensure that the project is completed successfully. There are several process models to select from, as the need be. Each process model has its strength in a given development scenario. E.g Sequential, Prototyping, Rapid application and so on.

Project Dimensions-cont'd

- **Product(Software):** This refers to the software that are being developed or produced. It is the scope of the software that is proposed to solve the requirements of the user. The software and its scope are a product that is defined and described by four factors namely, context, objective, functions and performance.
- A software product may fall into any one of the following categories:
 - a) System software
 - b) Application software
 - c) Software development tools
 - d) Software solution
 - e) Software generic product for general requirements
 - f) Software tools: automation, printing, processing, report writing, screen generation and so on.

Project Dimensions-cont'd

- **Technology:** It plays a critical role in SPM as it can help improve the efficiency and effectiveness of the project. It involves language and tool selection, value and cost of reuse.
- Here are some examples of technology that can be used in SPM:
 - Project management software e.g. Trello, Asana, and JIRA
 - Communication tools: e.g. Slack, Zoom, and Microsoft Teams.
 - Version control software: e.g. Git and subversion
 - Automated testing tools: e.g. Selenium and JMeter.
 - Cloud computing: e.g. Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform.

Software Project Management

- It is the collection of techniques used to develop and deliver various types of software products.
- It involves a range of tasks that ensure the successful completion of software projects. Some of the key tasks include:
- **Project Planning:** This involves defining the project scope, setting project goals, creating a project plan, and developing a timeline for project completion.

Software Project Management Tasks

- **Resource Management:** This involves identifying the resources required to complete the project, assigning roles and responsibilities to team members, and ensuring that the required resources are available when needed.
- **Risk Management:** This involves identifying potential risks that may impact the project, assessing the probability and impact of each risk, and developing strategies to mitigate or avoid the risks.
- **Quality Management:** This involves defining quality standards for the project, ensuring that the project meets these standards, and conducting regular quality reviews to identify and address any issues.

Software Project Management Tasks

- **Communication Management:** This involves establishing communication channels and protocols for the project, ensuring that team members are informed of project progress, and communicating with stakeholders to keep them informed of project status.
- **Schedule Management:** This involves monitoring project progress against the project plan, identifying any schedule deviations, and developing strategies to bring the project back on track if necessary.
- **Budget Management:** This involves tracking project expenses against the project budget, identifying any variances, and taking corrective action if necessary.

Software Project Management Tasks

- **Change Management:** This involves managing changes to the project scope, requirements, or schedule, and ensuring that any changes are properly documented and communicated to stakeholders.
- **Team Management:** This involves managing team dynamics, providing feedback and coaching to team members, and resolving conflicts that may arise.
- **Stakeholder Management:** This involves identifying project stakeholders, understanding their needs and expectations, and managing their involvement in the project to ensure their satisfaction with the project outcome.

Software project **vs.** others

Are *software* projects really different from other projects?

- **Invisibility**: progress in software project is not immediately visible.
- **Complexity**
- **Conformity**: software system has to conform to the requirement of human clients.
- **Flexibility**: one strength of software is its flexible and easy to change.

Problems with software projects

commonly experienced problems (from the manager's point of view)

- poor estimates and plans;
- lack of quality standards and measures;
- lack of guidance about making organizational decisions;
- lack of techniques to make progress visible;
- poor role definition – who does what?
- incorrect success criteria.

Problems with software projects (Another point of view)

- lack of knowledge of application area;
- lack of up-to-date documentation;
- preceding activities not completed on time;
- lack of communication between users and technicians;
- lack of commitment – especially when a project is tied to one person who then moves;
- narrow scope of technical expertise;
- changing requirements;
- changing software environment;
- deadline pressure;
- lack of training